Angle of incidence: The angle formed between the incident ray PO and the normal 'ON' is angle of incidence. It is denoted by $\angle i$

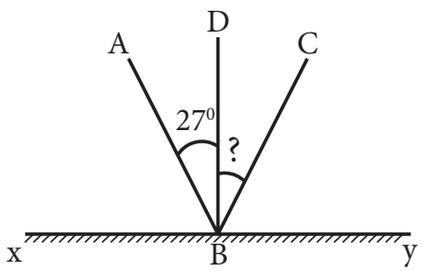
Angle of reflection: The angle formed between the reflected ray OQ and the normal ON is angle of reflection. It is denoted by $\angle r$

Laws of reflection:

1. The angle of incidence is always equal to the angle of reflection. $\angle i = \angle r$
2. The incident ray, the reflected ray and the normal at the point of incidence lie on the same plane.

Example 1

In the figure, the incident ray makes 27° with the normal, then find the angle of reflection.



Solution:

Angle of incidence = 27°

According to the laws of reflection, the angle of reflection = Angle of incidence = 27°

Example 2:

A light ray strikes a reflective plane surface at an angle of 43° with the plane surface.

i. Find the angle of incidence.

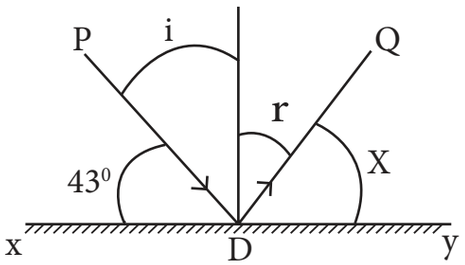
ii. Find the angle of reflection.

iii. Find the angle between the incident and the reflected ray

iv. Find the angle between the reflected ray and the plane surface.

Solution:

We use the diagram shown below to answer the questions.



a) Angle of incidence $\angle i = 90 - 43 = 47°$

b) angle of reflection $\angle r = \angle i = 47°$

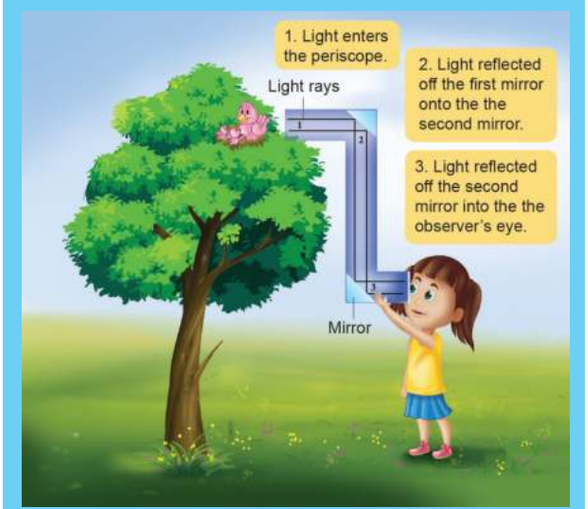
c) $\angle i + \angle r = 47 + 47 = 94°$

d) $\angle x = 90 - \angle r = 90 - 47 = 43°$

ACTIVITY 3

Make your own Periscope : You can use an empty agarbathi box and two plane mirrors to make a periscope.

As shown in the figure below, two plane mirrors are kept 45 degree to horizontal.



As shown the figure above, the light rays from the distant object enter through the tube at 1, and hit the mirror at 2. As the angle of incident must be equal to angle of reflection, the reflected rays flow through the tube downwards. As the light rays hit the mirror at 3 once again they are reflected. This reflected rays then travel out of the box to our eye. As you can see, periscope uses the laws of reflection.

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